

Restoration of Chromium Contaminated Soil-Groundwater Systems at COPR Site in Rania and Khan Chandpur Villages, India

Field Work Plan 2024-25



by

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1. Introduction

Observing the rapidly growing concern over the groundwater contamination and deteriorating soil-water quality, we moved to Rania and Khan Chandpur villages in Kanpur Dehat district in Uttar Pradesh. Rania-Khan Chandpur site is one of the highly contaminated sites in India (Prakash et al. 2011; CPCB, 2023) due to Chromite Ore Processing Residue (COPR), which is a hazardous (waste) remnant of the chromate (CrO_4^{2-}) extraction roasting process (Földi et al. 2013; Matern et al. 2017; 2020). As per report of the oversight committee, Hon'ble NGT, U.P. Lucknow, there was an estimated volume of 1,22,800 m^3 COPR at Rania site. Both the villages are settled in the vicinity of the leather processing clusters and COPR waste from these tanneries and related industries has severely contaminated soils, the surface-water, and the groundwater. The residents are drinking Cr-containing groundwater and are always at risk through the poisonous intake either through groundwater or locally grown vegetables or other sources (Guleria et al., 2022). Regardless of this situation, the remediation praxis of Cr-contaminated soil-water systems remain unattempted at the Rania-Khan Chandpur, this is because the site-scale Cr fate and mobility in the subsurface is still less investigated.

Cr leachate from the COPR can infiltrate in the vadose zone and ultimately meet the (shallow) groundwater which can result as huge Cr contamination in the subsurface. Our initial (Gupta et al. 2023a; Deoli et al. 2023; Gupta et al. 2024) and other works (Singh et al., 2009; Matern et al., 2017) have found that the groundwater, surface-water, and soil samples from the past have shown signs of a Cr emanating from the COPR dump site in and around Rania-Khan Chandpur villages. In the subsurface, Cr can be in the form of Cr (III) or Cr (VI), the mechanisms and kinetics of Cr release are obscure (Bhattacharya et al., 2019). The release of Cr from the COPR dump site can be influenced by factors such as the pH of the water/soil, the presence of other chemicals or ions, the hydrogeology and hydrology of the site, the above and below-ground ecology and microbial community dynamics, and so on. Further, the climate change may influence the fate and mobility of the Cr by altering the hydro-meteorological and the subsurface conditions. A better understanding is needed to improve remediation practices including evaluation of potential Cr transport pathways and rates, and more efficient clean-up

strategies. However, there remains a dearth of knowledge on Cr fate, transport, and remediation in the soil-water system.

Considering the research gaps prior to the implementation of this research work, a series of investigations has been initiated to generate baseline data of soil-water quality at the Rania-Khan Chandpur site. In this research work, the fate and mobility of Cr in the subsurface under varying hydrogeological conditions will be investigated first. Thereafter, the performance of the remediation techniques will be evaluated for restoration of Cr-contaminated soil-water systems in Rania-Khan Chandpur villages. This research will provide very accurate, multimethod techniques for the forecasting of the Cr mobility in the subsurface, which can assist the residents/ stakeholders for the future site scale planning. Further, this research will not only improve our understanding to manage a highly contaminated site in India, but also helps in management of water resources in rural India.

2. Literature Review

2.1 Global Scenario

About ten million tons of chromium are produced globally each year. South Africa, Turkey, China, India, and Kazakhstan are the top five countries for chromium production in the world from which South Africa produces almost two-fifths of the chromite ores and concentrates. (Mohanty et al., 2011; Mishra and Bharagave 2016). Countries like USA, Argentina, Pakistan, Bangladesh, Germany, Indonesia have suffered from Cr concentration in groundwater (Ali et al., 2022; Khan et al., 2010; Vainshtein et al., 2003; Astuti et al., 2023).

2.2 Indian Scenario

In India, there are more than 20 large Cr contaminant sites in which Rania-Khan Chandpur (Kanpur, Uttar Pradesh), Sukinda Valley (Odisha) and Ranipet (Tamil Nadu) are the worst-hit states with a high abundance of Cr in soil-water system (CPCB, 2023) and 112 probably Cr contaminated sites. In many other states, the levels of Cr have exceeded the permissible limit which creates the consequences. The highest concentration of Cr and Cr (VI) have been reported in Sukinda Valley with values of 48.7–250.2 and 21.4–115.2 $\mu\text{g L}^{-1}$ respectively (Naz et al., 2016).

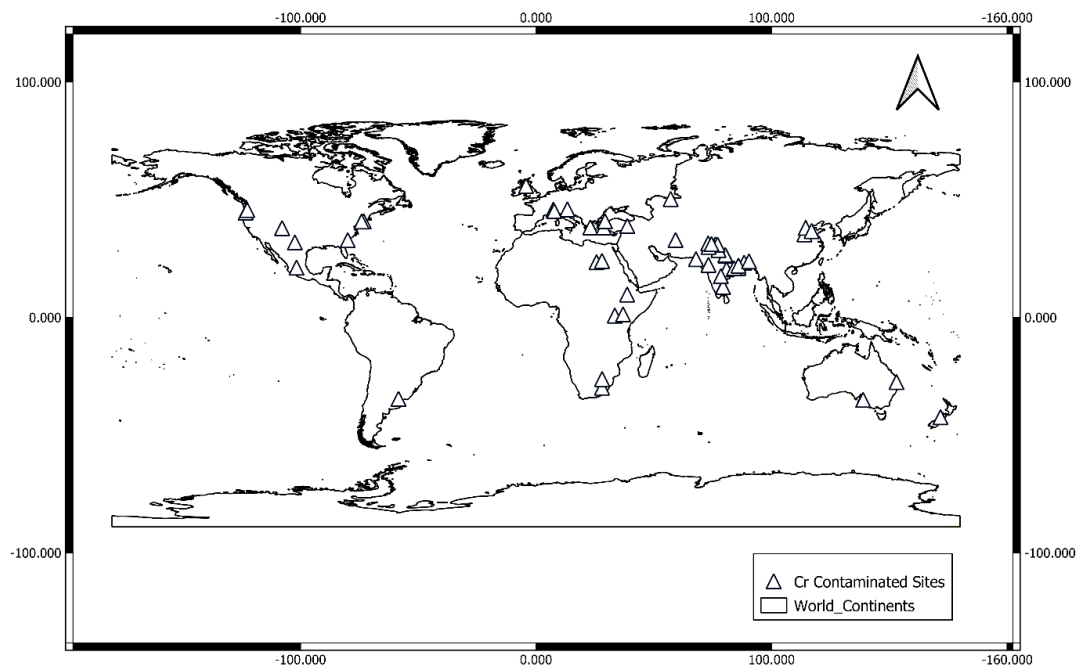


Fig. 1: Cr contaminated sites of world due to leaching of COPR

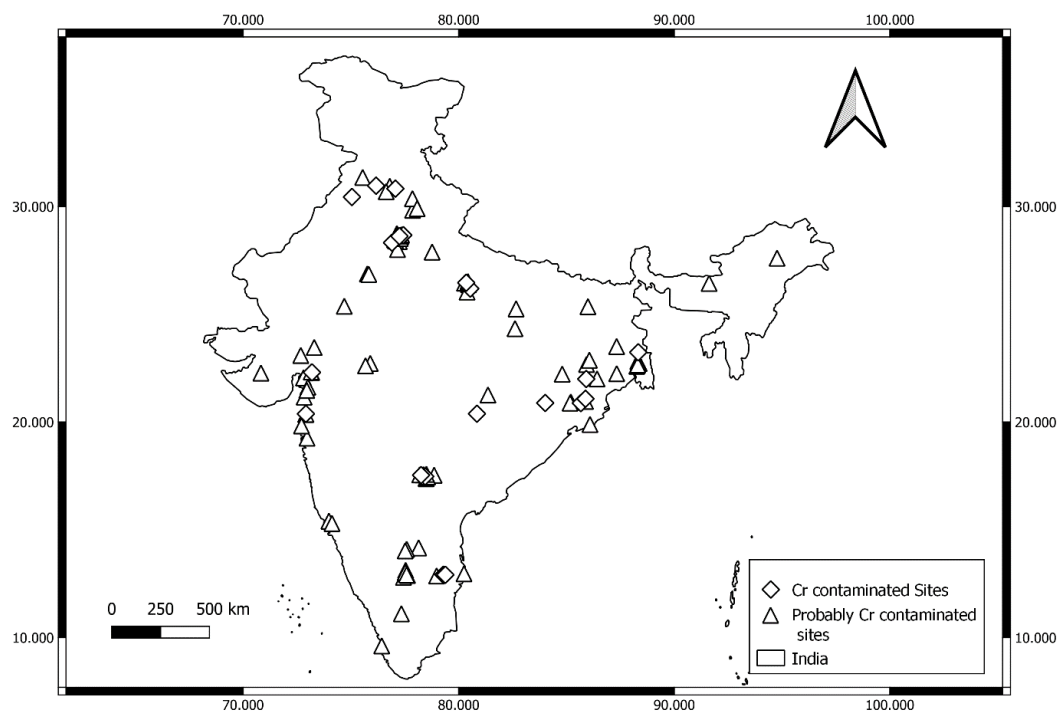


Fig 2: Cr contaminated and Probably Cr contaminated sites in India

2.3 Soil Water Quality at Cr Contaminated Soil-groundwater System

Due to the continuous leaching of Cr from COPR in the subsurface, many areas around the world were severely affected by Cr. In the past various studies were conducted for the contamination due to Cr (Table 1). Farmer et al. (2002) studied Cr in groundwater at Cr contaminated site Glasgow and found max concentration up to 91 mg L⁻¹ in groundwater and 6.7 mg L⁻¹ in surface water. Similar study has been conducted by Naz et al. (2006) at Sukinda Chromite Mine, Odisha and found max Cr concentration up to 250.2 mg L⁻¹. In Rania, Chhiwali and Godhrauli villages, Gularia et al. (2022) collected groundwater samples for health risk analysis and found maximum Cr concentration 115 mg L⁻¹. In our study, we collected groundwater samples in Rania and Khan Chandpur villages for all three seasons namely pre-monsoon, post monsoon and monsoon to analysis overall groundwater quality of the Cr contaminated area. Furthermore, the soil samples and cores have been collected from the study area to know the overall Cr contamination in COPR, soil as well as agriculture land at different depth.

2.4 Fate and Transport of Chromium in Soil-Groundwater Systems

2.4.1 Adsorption

The adsorption of Cr into soil-groundwater systems results in a reduction of concentration in the aqueous phase and a retardation of the velocity of Cr movement. Adsorption is a phenomenon where the accumulation and concentration of Cr at the interface of a sorbent material (Toth 2002). During adsorption, Cr moves from the bulk phase and adheres to the porous surface in a semi-liquid state, and this process is primarily driven by the concentration-to-solubility ratio of the compound. Table 2 shows the different adsorption studies in fate and transport of Cr. Jardin et al (2002) studied adsorption in lab scale study using K₂CrO₄ as a Cr. Ye et al. (2019) studied Cr (VI) adsorption by placing 1 g of soil in 50 ml of water. They found Soil-leaching solution that achieved Cr (VI) concentration of 905 mg/L, showing excellent fit with the pseudo-second-order kinetic model for adsorption behaviour. Cao et al. (2021) studied adsorption of Cr for West of Songnen Plain, China. Chromium (VI) solutions were prepared with K₂Cr₂O₇. In our study, for adsorption of Cr, batch experiments will be done in different combinations of pH and water content. The pH will be varied as 5,7, 9 and 11 with water content 0.8, 1 and 1.2. It will be conducted on the soil collected from the Rania and Khan Chandpur and Cr contaminated groundwater will be used as a Cr source.

Table 1: Scenario of soil-groundwater quality at COPR sites

Citation	Objective	Study Area	Material	Max Cr concentration	Key Observation
Farmer et al. (2002)	To determine Cr contamination in groundwater and surface water	Glasgow	Groundwater	91 mg L ⁻¹ in groundwater and 6.7 mg L ⁻¹ in surface water.	Site specific conditions (pH, Temperature, Rainfall, organic matter) play an important role in Cr dissolution.
Guleria et al. (2022)	To determine the ecological risk, human health risk in groundwater	Rania, Chhiwali and Godhrauli	Groundwater	115 mg L ⁻¹	Rania soil contact: >1–2 order higher cancer risk; Chhiwali groundwater: CR = 18.5 and 54.9 (>10 ⁻⁶) for adults.
Naz et al. (2006)	To determine the human health risk of Cr (VI) and Cr (III) via groundwater.	Sukinda Chromite Mine, Odisha	Groundwater	250.2 mg L ⁻¹	The total cumulative average cancer risk and non-cancer risk (Hazard Quotient) was found 2.04E-03 and 1.37 in male and 1.73E-03 and 1.16 in the female population
Eliopoulos and Megremi (2021)	Status of groundwater for human health	Neogene Assopos and Thiva basins	Soil and Groundwater	40570 mg Kg ⁻¹ in soil and 21 mg L ⁻¹ in groundwater	High Cr concentration has been found in soil-groundwater system of the basin due to mining and smelting.
Driscoll et al. (2010)	Study of Cr contamination in COPR	New Jersey US	COPR	3000 mg Kg ⁻¹	High Cr concentration found in and around COPR dump.
Huang et al. (2009)	To determine the Cr contamination in COPR dump	Hunana, China	Soil	2130.3 mg Kg ⁻¹	High Cr contamination in topsoil layers due to COPR leachate from industries
Anduyappan et al. (2017)	To determine the Cr concentration in agriculture land	Ranipet, India	Agriculture soil	79.71 mg Kg ⁻¹	The Cr concentration is more than the permissible limit in agriculture field
Kumar et al. (2023)	To determine the Cr in COPR at different depth	Rania, India	COPR	42570 mg Kg ⁻¹	Cr concentration increase with increase in depth.

Table 2: Different studies conducted on adsorption of Cr using batch and column experiment at different sites around the world

Citation	Objectives	Study Area	Experimental Conditions and Methods	Adsorption Coefficient value	Key Observation
Ye et al. (2019)	Study soil Cr (VI) adsorption, develop adsorption models, explore Cr (VI) transport, and predict Cr (VI) leaching from chromium slag in soil	Xiangxiang (112°28'42.91 "N, 27°44'50.96" E)	Scale: Batch scale and column experiment (40 cm high and 7 cm diameter) Models: Kinetic models and Isotherm	$K_d = 1.17 \text{ cm}^3/\text{g}$ $R_d = 10.6$	Soil-leaching solution achieved Cr (VI) concentration of 905 mg/L, showing excellent fit with the pseudo-second-order kinetic model for adsorption behavior.
Khan et al. (2010)	Develop models to understand hexavalent chromium sorption and transport in soil media.		Scale: Column experiment Analytical Method: UV spectrophotometer	$K_d = 0.0115 \text{ cm}^3 \text{ g}^{-1}$ $R_d = 2$	Initial Cr (VI) concentration notably affects equilibrium uptake. Low concentrations at cycle ends.
Wu et al. (2022)	Develop a mathematical model utilising data on Cr (VI) migration and leaching transport to forecast Cr pollution in the "slag-soil-groundwater" system.	Xiangxiang City, China	Scale: Column experiment (height 81 cm)	$K_d = 0.0542 \text{ cm}^3 \text{ g}^{-1}$ $R_d = 1.17$	The concentration of leached Cr (VI) and Cr (VI) slag increases.
Yan et al. (2023)	To understand Cr migration and its interaction with different soil	Shijiazhuang, Zhuzhou and Guangzhou	Scale: Column experiment (1.2 cm diameter and 8 cm high)	$K_d = 0.84 \text{ cm}^2 \text{ g}^{-1}$ $R_d = 1.43$	The soil of Zhuzhou had the maximum adsorption capacity for Cr (VI) (0.225mol/kg).

	types, and the associated long-term risks in studied sites.	China,	Analytical Method: UV spectrophotometer		Rainfall intensity impacted Cr (VI).
Arthur et al. (2017)	Examine DNAN fate in soils, quantify reversible adsorption, assess mass loss, and analyse DNAN-soil property relationships.	Entisols, Inceptisols, Mollisols, Aridisols, USA	Scale: Batch adsorption Models: Freundlich and linear isotherms	$K_d = 0.0016, 0.0051$ $R_d = 3.5$ to 27.5	The equilibrium adsorption data were fitted to Freundlich and linear isotherms with the Freundlich isotherm resulting in slightly better fit on average.
Jardine et al (2011)	To investigate the fate and transport of Cr (III) and Cr (VI) in three different soil types	South Carolina and Hanford facility Washington	Scale: Batch and column Analytical method: Model: Freundlich isotherms	$K_d = 0.2 \text{ cm}^3 \text{ g}^{-1}$ and $14 \text{ cm}^3 \text{ g}^{-1}$	Contaminant transport depends on sorption linearity, kinetics, and soil properties influencing Cr-solid interactions.
Cao et al. (2021)	to evaluate the adsorption and transport of Cr (VI) on several natural sediments	West of Songnen Plain, China	Scale: Batch scale and column experiment Models: Kinetic models and Equilibrium adsorption isotherms	$K_d = 0.897$ to 6.430 L kg^{-1} $R_d = 2.86$ to 71	The order of adsorption capacity in different soil was loam, silty loam clay, sandy loam, and loamy sand, respectively. The adsorption of Cr (VI) decreases with increase of the pH and solid/solution ratio.

2.4.2 Leachability

The Cr compounds (Cr^{3+} and Cr^{6+}) that leach through groundwater are dependent upon the effluent's presence, gradient flow direction, and beginning concentration. The main forms of Cr (VI) are bichromate (HCrO_4^-) and chromate (CrO_4^{2-}), which are highly soluble in most environmental pH ranges of 6.5 to 7.5. Kanchinadham et al. (2015) studied leachability of Cr contaminated soil in 3 cm diameter and 5 cm height block. Semi-dynamic long term leachability test was used to determine the leachability index. Zhang et al. (2021) studied leachability of Cr in different grain sizes of contaminated soil. The contaminated soil was divided into bulk soil, coarse soil, medium sand, fine sand, and silt-clay. The proportions of fine sand (43.25%) and silt clay (43.26%) exhibited greater leachability of chromium compared to bulk soil (32.04%). Furthermore, compared to bulk soil (68.77%), the leachability of Cr (VI) was higher in silt-clay (85.72%) and fine sand (75.19%). Hence, the particle size decreased, Cr and Cr (VI) leaching concentrations increased. In our study, the column experiment will be an analysis for leachability of soil-COPR cores of Rania and Khan Chandpur villages.

2.5 Transport of Chromium

The Cr, which primarily occurs as Cr (III) and Cr (VI), exhibit different transport behaviours in soil-groundwater systems. Cr (III) has a propensity to adhere firmly to soil particles and create insoluble precipitates, which reduces its mobility. The pH, organic matter content, and mineral composition of the soil are important factors that affect Cr (III) transport. On the other hand, Cr (VI), which is more soluble, is vulnerable to redox changes, and the pH and prevailing redox conditions have an impact on its mobility. Advection, dispersion, sorption, and degradation are some of the physical and biological mechanisms that cause chromium to migrate through soil. The two primary mechanisms that allow Cr (VI) to transport from the subsurface to the ground surface are capillary rise and evaporative enrichment, and evaporative enrichment (Henry et al., 2007). Contaminants may rise to the surface through capillary action when the capillary rise is higher than the distance between the water table and the surface (Henry et al., 2007; Hillel, 1998).

2.5.1 Advection

Advection is the movement of dissolved Cr with the underlying groundwater flow in soil-groundwater system. The amount of Cr being transported is a function of its concentration in the groundwater and the groundwater velocity. The behaviour of chromium during advection

is influenced by its speciation, with Cr (VI) exhibiting higher mobility due to its lower propensity for adsorption to soil particles compared to trivalent chromium. Chen et al. (2021) found that advection is the dominant transport mode that highly controls the Cr transport. Soils with lower permeability, including montmorillonite and kaolinite, show significant delays in the transfer of Cr. High montmorillonite content causes the slowest Cr movement and the lowest peak concentration in simulations, as it does in natural clay and montmorillonite soils.

2.5.2 Diffusion

Diffusion is the movement of a Cr in water from an area of higher concentration to an area of lower concentration. Even in the absence of fluid movement, diffusion will take place as long as there is a concentration gradient. If each adjacent layer of soil-groundwater system has the necessary porosity, diffusion can also occur when a contaminant's concentration is higher in one stratum than in another. The rate of diffusion (Q_d) at rest is given by Fick's Law:

$$q_d = -D_e \frac{\partial C}{\partial x}$$

Where D_e (m^2s^{-1}) is the coefficient of diffusion and C ($kg\ m^{-3}$) is solute concentration in soil-groundwater system. The minus sign shows the direction of diffusion is opposite to the direction of increasing concentration. Diffusion coefficient in soil-groundwater system is determined by mathematical analysis as a function of position or time.

2.5.3 Dispersion

Dispersion is a process which spreads the solute in soil-groundwater systems due to groundwater velocity difference in different direction/flow path. Differences in the flow velocities due to the different pore size and shape (area and volume of the solute) cause solute transport at different rates and thus lead to dispersion or mixing of an incoming solution. The different dispersion studies on Cr are shown in table 3. Furthermore, the dispersion may occur in soil-groundwater system due to the development of a velocity profile within an individual pore such as the highest velocity occurs in centre and no flow or negligible flow at the walls, different mean velocities of different size of pore, the mean water flow direction being different from the actual streamlines in the soil-groundwater system and solute particles diverging or converging from the same pore.

The solute flux is given by an equation below.

$$q_h = -D_h \frac{\partial C}{\partial x}$$

Where q_h (m/s) is the solute dispersive flux D_h dispersion coefficient ($m^2 s^{-1}$), C ($kg m^{-3}$) is solute concentration in soil-groundwater system.

In our study, a 3D tank will be used to transport of Cr in soil-groundwater system of Rania which will be the first study of Cr transport in soil core. In this study the core will collected from the field directly with help of core cutter. The water movement in the tank's setups will be modelled using the Richard's equation with HYDRUS software.

2.6 Remediation of Cr contamination

Soil-water systems can potentially contain elevated concentrations of chromium (Cr), which can manifest as a significant threat due to its potential for inducing severe toxicity in humans, plants, and animals (Escudero et al. 2017). Therefore, the remediation of soil contaminated with Cr is important for clean drinking water. Numerous remediation strategies have been developed with the primary objective of mitigating chromium (Cr) contamination risks, diminishing land fragmentation, facilitating land reclamation, and thereby fostering the expansion of agricultural land to ensure long-term sustainability. Jia et al. (2022) developed an improved indicator system for phytoremediator screening strategy and identified *Pennisetum* is an effective phytoremediator for Cr. Sajad et al. (2020) analysed 61 plants from 30 different families for concentration of Cr. Based on Bioconcentration Factor (BCF) value, 38 plant species show viability for the phyto-stabilization of Cr excluders. As indicated by Saravanan et al. (2019), this method stands out as one of the best approaches compared to conventional methods for remediating contaminated sites, as it leverages a diverse range of plant species for effective treatment.

Table 3: Different studies conducted on dispersion of Cr transport in soil-groundwater system at sites around the world

Citation	Objectives	Study Area	Experimental Conditions and Methods	Dispersion Coff value	Key Observation
Yan et al. (2023)	To study the fate and transport of Cr in soil	Shijiazhuang, Zhuzhou and Guangzhou, (China)	Scale: Column experiment Model: Saturated water flow model, Solute transport model,	$D = 7.84 \text{ cm}^2 \text{ h}^{-1}$	the horizontal transport of pollutants caused by groundwater will also lead to the diffusion of heavy metal.
Trento and Alvarez (2011)	Examine the fate of Cr in soil and evaluate the risks of vertical motion.		Scale: Column Models: Mathematical model, Numerical model	$D = 10.6 \text{ m}^2 \text{ cm}^{-1}$	The Salado River is fully affected by Chromium. An accurate approximation of chromium concentrations in water and bed sediments.
Khan et al. (2010)	To model the transport of Cr (VI) through soil and estimated the time for reaching the contaminant in layers of aquifers		Scale: Column experiment Models: Adsorption isotherm, Freundlich isotherm, Langmuir adsorption isotherm	$D = 0.00475 \text{ cm}^2 \text{ h}^{-1}$, $0.00732 \text{ cm}^2 \text{ h}^{-1}$ and 0.00768 at 1.2 ml min^{-1} , 1.0 ml min^{-1} and 0.8 ml min^{-1} flow rate	The initial concentration of Cr (VI) has a substantial effect on equilibrium absorption. Low concentrations at ends are caused by the inorganic pollutant concentration.
Selim et al. (1989)	To investigate the mobility of Cr (VI) in		Models: Kinetic model, Convective-dispersive transport equation,	$D = 0.0001296$ to $0.0004464 \text{ m}^2 \text{ d}^{-1}$	All soils' predictions of the Cr breakthrough curve were improved by optimised

	six soils with different properties				parameters using a nonlinear least squares approach.
Boupha et al. (2004)	To develop a software library and user interface that provide solutions to contaminant transport equations to invert Laplace transforms		Models: Laplace transforms, The De Hoog algorithm	$D = 0.2 \text{ m}^2 \text{ d}^{-1}$	The purpose of the program is to readily provide solutions to encourage a better understanding of factors affecting the fate and transport of groundwater contaminants.
Yolcubal and Akyol (2007)	To determine the sorption & transport of Cr (VI) in a high alkaline and calcareous soil.	Antalya, Turkish Mediterranean coast	Scale: Batch experiment and Column experiment	$D = 44.7 \text{ cm}^2 \text{ h}^{-1}$ (saturated experiment) and $0.3 \text{ cm}^2 \text{ h}^{-1}$ in unsaturated condition	Cr (VI) transport was marginally delayed compared to nonreactive tracer at both saturated and the high mobility of Cr (VI) enhances the rate of pollutant distribution

3. Research Gaps:

Soil-groundwater quality management is lacking in Rania and Khan Chandpur villages. A little information is available on the Cr concentration in the soil-groundwater system (Matern et al. 2017; Mishra et al. 2019; Kumar et al., 2023). Therefore, this research will investigate the site-scale Cr mobility in the subsurface under varying hydrogeological conditions. We are not aware about the status of Cr contaminated aquifers from where drinking water supply. This study will do a complete hydro-geological assessment of the site. Furthermore, the fate and transport study of Cr is completely lacking in this area which will be done in this study and then present and future contamination load will be determined. Due to high Cr contamination the design of remediation techniques is required for Cr contamination removal from soil-groundwater systems which helps to improve clean up strategies for soil-water systems.

4. Objectives of the Study

1. To investigate the soil-groundwater quality in Rania and Khan Chandpur villages and conduct a survey by questionnaire and capacity building programmes.
2. To investigate the fate and transport of Cr in soil-groundwater systems under current and future hydrological site conditions in and around Rania-Khan Chandpur villages.
3. To design remediation techniques for Cr contaminated soil-groundwater systems and evaluate the performance of such techniques including PRB for Cr removal in Rania-Khan Chandpur villages.

5. Study Area

Kanpur Dehat district has an area of about 3021 km² and the 2 dumpsites located at Rania-Khan Chandpur near NH-2 Toll Plaza 10 km east from the main densely populated city was Akbarpur which was approximately 20 km west side from Kanpur city. Rania-Khan Chandpur site is one of the highly contaminated sites in India and there is an estimated volume of 1,22,800 m³ COPR at Rania site. This region has humid subtropical climate conditions with average rainfall 782.8 mm (Srivastava et al., 2013) from which About 70–80% of the total precipitation occurs in July, August, and September during the monsoon season (Matern et al., 2017).

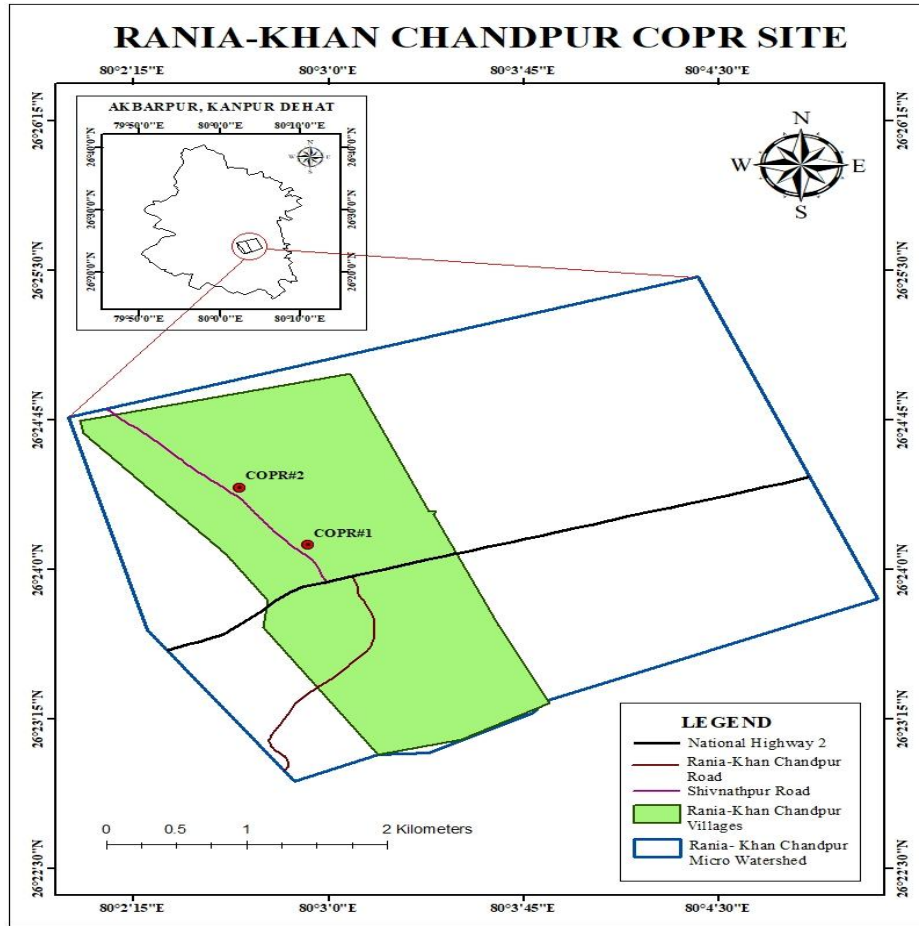


Fig 3: Map of Rania and Khan Chandpur Villages with two COPR dump sites as COPR#1 and COPR#2. COPR#1 is the main dump site and COPR#2 is the relevantly small dump side in north of the COPR#1. National Highway (NH 2) separated the Rania and Khan Chandpur villages as in north side Rania Village and Khan Chandpur is in south direction

6. Research Methodology

The research aimed to perform village-scale assessment of soil-water resources in Rania and Khan Chandpur villages. Furthermore, this research will develop a multiscale approach to investigate the fate and transport of Cr in soil-groundwater systems. The entire investigation is projected in three major phases, in the first phase the village level intervention will be performed to investigate spatial distribution of Cr in soil-groundwater systems followed by continuous monitoring using laboratory simulation. In the second phase, the field and modelling work will be finalized to upscale the fate and transport mechanisms under the real site hydrological and under the future climatic conditions. Later, based on the finding of the laboratory, field, and modelling works, the designs of the remediation techniques including PRB will be tested to develop restoration plans for Rania and Khan Chandpur villages.

Work Plan#1:

Groundwater Quality Assessment

A total of 78 groundwater samples were collected in the post-monsoon season in year 2022 while 55 groundwater samples were collected in monsoon season 2023 and 29 groundwater samples were collected in pre-monsoon season in year 2024 from Rania-Khan Chandpur villages to check the status of groundwater quality in and around the COPR sites. Heavy metal (Cr, Fe, K)) and chemical analysis (pH, EC, TDS, Total Hardness, Ca, Mg, Cl) have been performed for all the samples. Chemical analysis has been performed in Applied Microbiology Lab, CRDT, IIT Delhi and heavy metal analysis was done using MP-AES (Agilent Technologies 4200) in the same lab. The acid digestion was performed by 1 ml of HNO₃ added in 4 ml of water sample using Anton Paar microwave digestion system. Sample was digested for 30 minutes with maximum temperature of 180 °C.

Cr Distribution in Soil Profile

To know the Cr contamination within soil and COPR samples, undisturbed cores have been collected from different depths (0-5 cm, 50-55 cm, 100-105 cm, and 150-155 cm). The Cr analysis has been done using MP-AES. The acid digestion of samples collected from different depths was performed using aqua regia solution (HCl: HNO₃ :: 3 : 1) was performed. For this, 0.25 g of soil sample was mixed with 4 ml of aqua regia solution and acid digestion was done as described above for groundwater quality analysis.

A questionnaire will be prepared to collect information regarding soil-groundwater contamination from respondents. Methods such as questionnaire survey, interview with interest groups and expert consultation were adopted to analyse and evaluate the environmental awareness of residents in this basin, and the current situation of participation in water resource management in contaminated area. The research includes knowledge of soil-groundwater contamination mainly due to COPR sites in that area, environment knowledge of locals, environment satisfaction, information circulation channel, the willingness of the public to participate, actual participation, and so on.

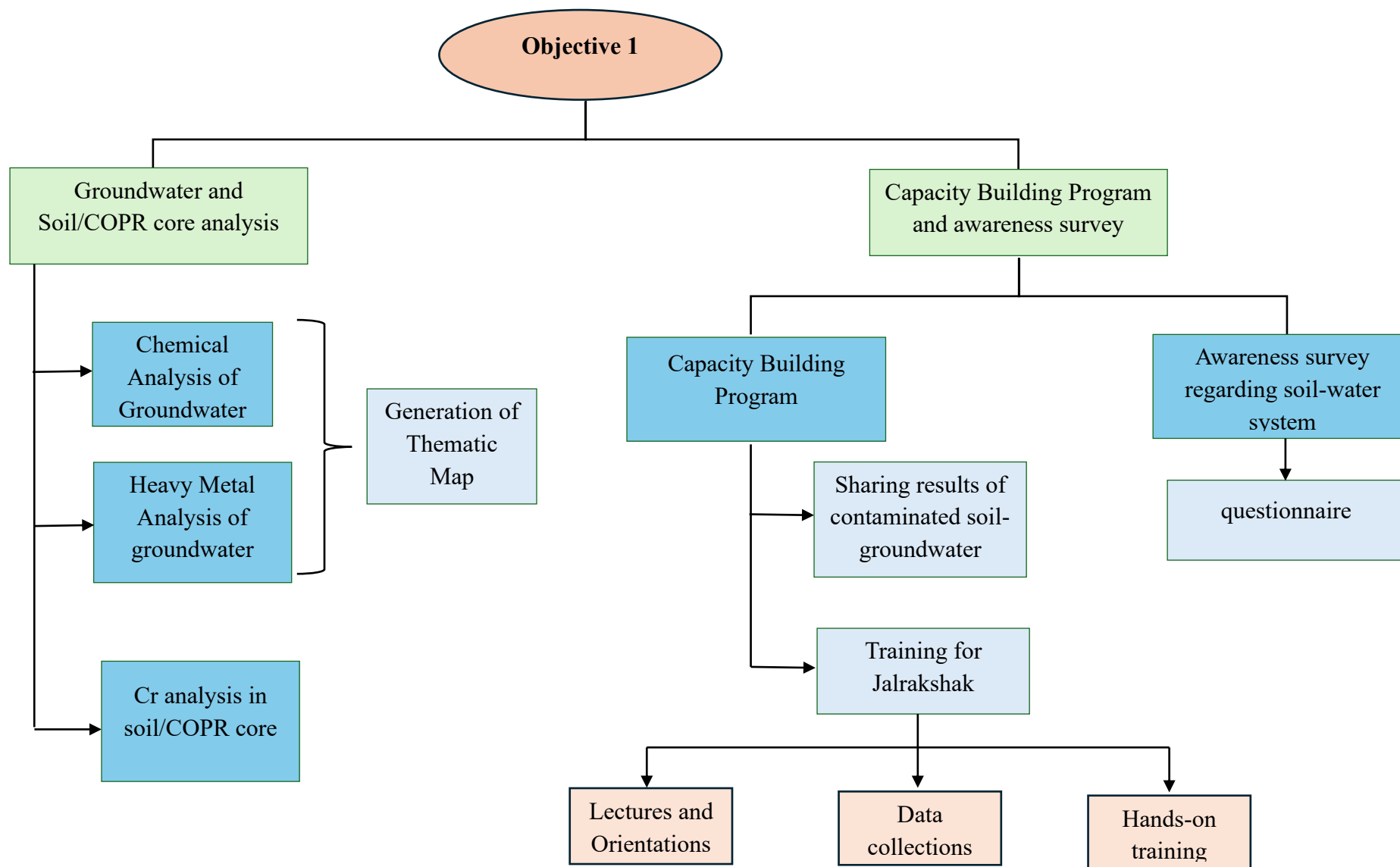


Fig. 4: Flow Chart for methodology adopted in groundwater, soil/COPR core analysis and capacity building program

Understanding the soil-groundwater contamination in rural areas is very limited and not properly reached to the grassroot level of village communities. The main aim of our Prof. Ram Chandra Capacity Building Program is to educate the villagers towards the soil-groundwater contamination and share the finding of soil-water quality with the local community. So that they can know the contaminated and the uncontaminated zone of Rania-Khan Chandpur which will help in drinking clean water. Prof. Ram Chandra capacity building program will train and engage the youth of Rania-Khan Chandpur in management of soil-groundwater systems, and they will be called “Jalrakshak” a Hindi word which means “Water Protector”. The training program for Jalrakshak will be given at various stages including lectures and orientations, data collection, testing of data, review, and verification. Post-training mentoring of Jalrakshak is a most important aspect. It involves regular check-ins, field visits, and skill refinement workshops to support Jalrakshaks in their role as Water Protectors. Through peer learning opportunities and access to resources, they can effectively address soil-groundwater contamination issues and engage with the local community for sustainable solutions.

Work Plan#2: Laboratory and numerical modelling to mimic Rania-Khan Chandpur site conditions:

A series of 2-D and 3-D scale experiment will be performed to mimic the real site hydrogeological conditions in 3-D tank to characterize the Cr fate and transport parameters using undisturbed cores collected from the Rania-/Khand Chandpur. Two-dimensional and three-dimensional tanks have already been fabricated as designed by Gupta et al. (2019) and Gupta and Yadav (2020), respectively. The main body is a constructed tank of one piece of 1.5 mm thick stainless steel formed into a box with inner dimensions of 100 cm-long $\times$ 60 cm-width $\times$ 45 cm-deep. The front cover is made of a thick sheet and the space between the two walls is packed with undisturbed cores. The groundwater will be flowing horizontally from left to right controlled by two end chambers. The spatial and the temporal distributions of the Cr and its concentration in soil-water phase along with soil moisture content will be observed using Inductively coupled plasma mass spectrometry (ICP-MS).

Initially, sample will be characterized for their hydro-physical (Porosity, Bulk Density, K_{sat} , VGM parameters, soil moisture retention, etc.) and chemical properties, (pH, organic matter content, Cr, and other hazardous materials through applications of synchrotron X-ray/optical methods). The collected cores will be transferred to the 2-D and 3-D glass tanks built which will be placed on a Lysimeter for water budget estimation and attached with auxiliary columns

to maintain the groundwater flow/fluctuation dynamics. Multiple monitoring wells will be installed at different length and depths to collect soil-water samples. Soil-water samples will be collected with the target time interval to quantify the Cr and co-pollutants concentrations.

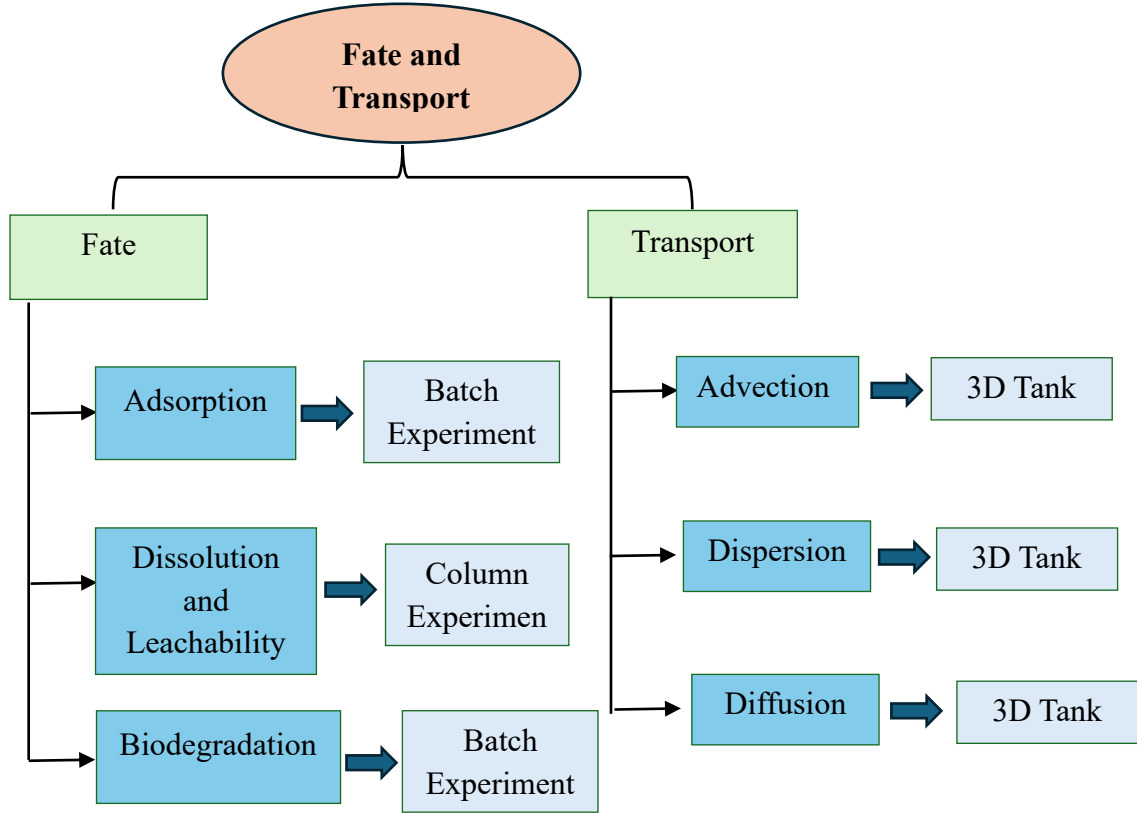


Fig 5: Flow Chart for Fate and Transport of Cr Contamination

The water movement in the tank's setups will be modelled using the Richard's equation with HYDRUS 2/3D software. In porous media, the flow and transport are modelled as nonlinear coupled partial differential equations. The equations of unsaturated flow and solute transport are as follows:

$$\frac{\partial \theta}{\partial t} - \frac{\partial K}{\partial \theta} \frac{\partial \theta}{\partial z} - \nabla \cdot (D_{\theta} \nabla \theta) = 0 \quad (1)$$

$$\frac{\partial (\theta C)}{\partial t} + q \cdot \nabla C - \nabla \cdot (D_C \nabla C) = R_C \quad (2)$$

Here $\theta(x,t)$ and $C(t,x)$ are respectively the moisture content and solute concentrations. Cr will be main solute, to be solved using reactive transport model and HYDRUS software. Both the unsaturated hydraulic conductivity $K(q)$ and moisture diffusivity $D_q(q)$ are nonlinear functions of q . The flux vector is the capillary pressure head. The reaction term in transport RC is often a nonlinear function of the solute concentration. During the multi-species transport process, C becomes a vector to represent the concentration of all species and reaction term becomes a function of all species concentrations. Initial and background conditions will be incorporated based on the observations from the current Cr concentration and hydrological conditions at Rania-Khand Chandpur site as time-variable boundary conditions..

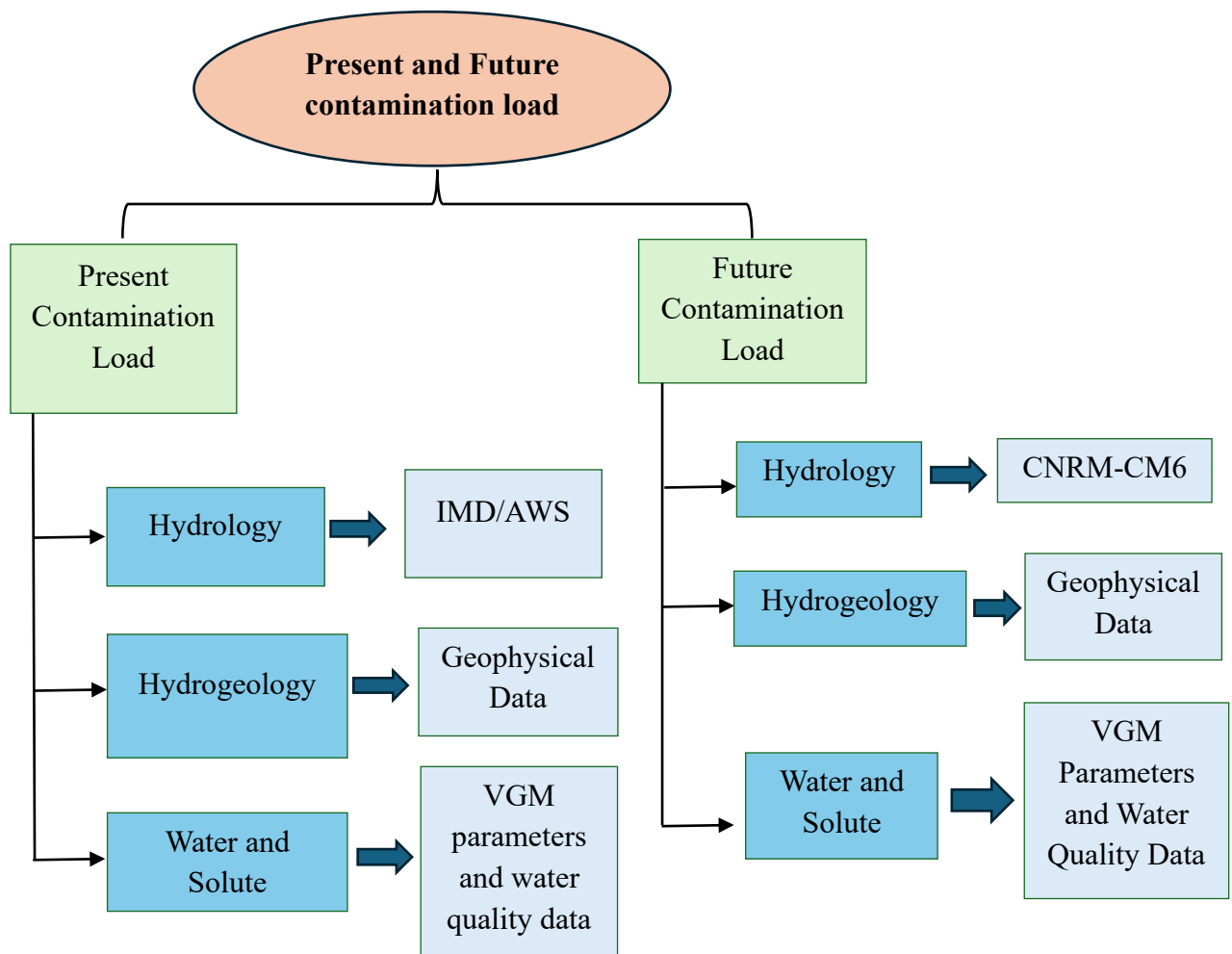


Fig 6: Flow Chart for determination of present and future contamination load

We will involve site-scale quantification of the hydrological fluxes and hydrogeology variables (for example, groundwater flow) to model Cr mobility in and around Rania-Khand Chandpur. First, the water budget component will be analysed by performing hydrological data collection

(precipitation (P), evapotranspiration (ET), etc.) to understand the dominating water fluxes (P vs ET) and to calculate the fluxes in/out from the site-scale soil-water system. A site-scale numerical domain will be built using MODFLOW (USGS[PG1]) to mimic hydrogeological formation of the Rania-Khand Chandpur using the geophysical data which has been already collected by my research group. The Cr mobility will be solved using the MODFLOW, in which the laboratory investigates the fate and transport parameters will be incorporated to run reactive flow of Cr. To validate the outcomes of modelling, the groundwater Cr concentrations and other geochemical variables will be used. Thereafter, the current and future climatic scenarios (data from CMIP5/CMIP6, <https://esgf-node.llnl.gov/projects/cmip6/>) will be added in the model to forecast the Cr contamination loads in the forthcoming years.

Outcomes of work #2: Soil moisture flow, Cr fate and transport parameters, model for upscaling the Cr mobility at site scale, a better understanding of the role of the subsurface conditions on Cr fate and transport. Cr fate and transport model to accurately predict and forecast Cr contamination loads at Rania-Khan Chandpur site.

Work Plan#3: Test for remediation of Cr and for performance evaluation of PRB at site:

Based on the outcomes of work plan#1 and work plan#2, first, the same MODFLOW model will be incorporated with different remediation conditions to explore (numerically) the possibility of a demonstrative scale test at a real site. In this process, highly reactive PRB material biochar will be placed in and around the Cr source zone under site hydrogeological conditions to optimize the design of the proposed PRB. Effectiveness of the PRB will be optimized using laboratory tested material. This numerical step will assist the demonstrative scale remediation of the Cr in soil-water system at Rania-Khan Chandpur. Finally, a series of demonstrative scale remediation activities will be implemented in Rania-Khan Chandpur and the effectiveness will be measured for the long term. The design, size, and other operational variables of such demonstrative scale remediation will be finalized based on the outcome of modelling works and will be improved time-to-time based on the post-implementation observations.

Outcomes of work plan#3: A multimethod approach to treat Cr contaminations at Rania-Khan Chandpur site.

Overall outcomes of the study

This work will help local rural communities to make informed decisions about drinking water resource management, land use management, and infrastructure development to minimize exposure to such hazardous contaminants. This will also support the implementation of low-cost commercial site scale measures to restore such a highly contaminated site, ensuring safe drinking water for the local population.

7. Timeline

S.N.	Activities	Timeframe (Semesters)							
		I	II	III	IV	V	VI	VII	VIII
1.	Coursework completion								
2.	Surface and Groundwater Quality analysis, Soil Core analysis								
3.	Batch and Column Study								
4.	3-D scale experiment to investigate the fate and transport of Cr in soil-groundwater systems and modelling								
6.	Cr mobility in the subsurface and forecast the future Cr-contamination loads								

7.	Design remediation techniques for Cr contaminated soil-groundwater systems								
8.	Synopsis and Thesis Submission								

8. Outcomes of the study and Target Beneficiaries

This work will help local rural communities to make informed decisions about drinking water resource management, land use management, and infrastructure development to minimize exposure to such hazardous contaminants.

The target beneficiaries of this research will be the local community living near the Rania COPR Site, who will be benefited from improved understanding of chromium contamination, potential health risks, and the implementation of effective restoration techniques. Furthermore, the regulatory/environmental authorities (like the CPCB/UPPCB) and other policymakers will benefit from the research results to develop strategies for contamination management and protection of soil-groundwater systems. Education and community involvement will be the exceptional outcome of this work. The scientific community will acquire important new information on the fate, transport, and remediation techniques of chromium, expanding their understanding of environmental remediation. Considering India has a growing environmental restoration marketplace of \$100 billion; this study will also help future startups to explore the commercial scale restoration and remediation technologies (Mahajan et al. 2022).

9. Work Done

9.1 Groundwater Quality Analysis

Groundwater in Rania-Khan Chandpur is mainly used for drinking, agriculture, and industry purposes. Therefore, the groundwater contamination in the study area seriously threatens the health of residents. The concentration of Cr is much higher than the permissible drinking limit (0.05 mg L^{-1}) in Rania-Khan Chandpur. As compared to Rania ($0 - 10.45 \text{ mg L}^{-1}$) Cr

concentration is higher in the Khan-Chandpur Village (0 - 46 mg L⁻¹) which is situated downstream of the COPR dump sites. The maximum Cr concentration has been found 46 mg L⁻¹ in post-monsoon season, Oct 2022 (Fig. 7), followed by monsoon season (39.48 mg L⁻¹), July 2023 (Fig. 8). Samling of pre-monsoon season has been done in February 2024 and the max Cr contamination was 27.30 mg L⁻¹ (Fig. 9).

We have conducted Prof. Ram Chandra Memorial Capacity Building Program on Water Resources Management in Rania-Khan Chandpur in a significant stride towards addressing water resources management challenges in rural areas on 13 to 15 February 2024 (Fig. 10). This program aims to share the results of water quality with local people. Moreover, the program served as a platform for fostering collaboration and partnerships among various stakeholders, including government agencies, academic institutions, NGOs, and local community leaders.

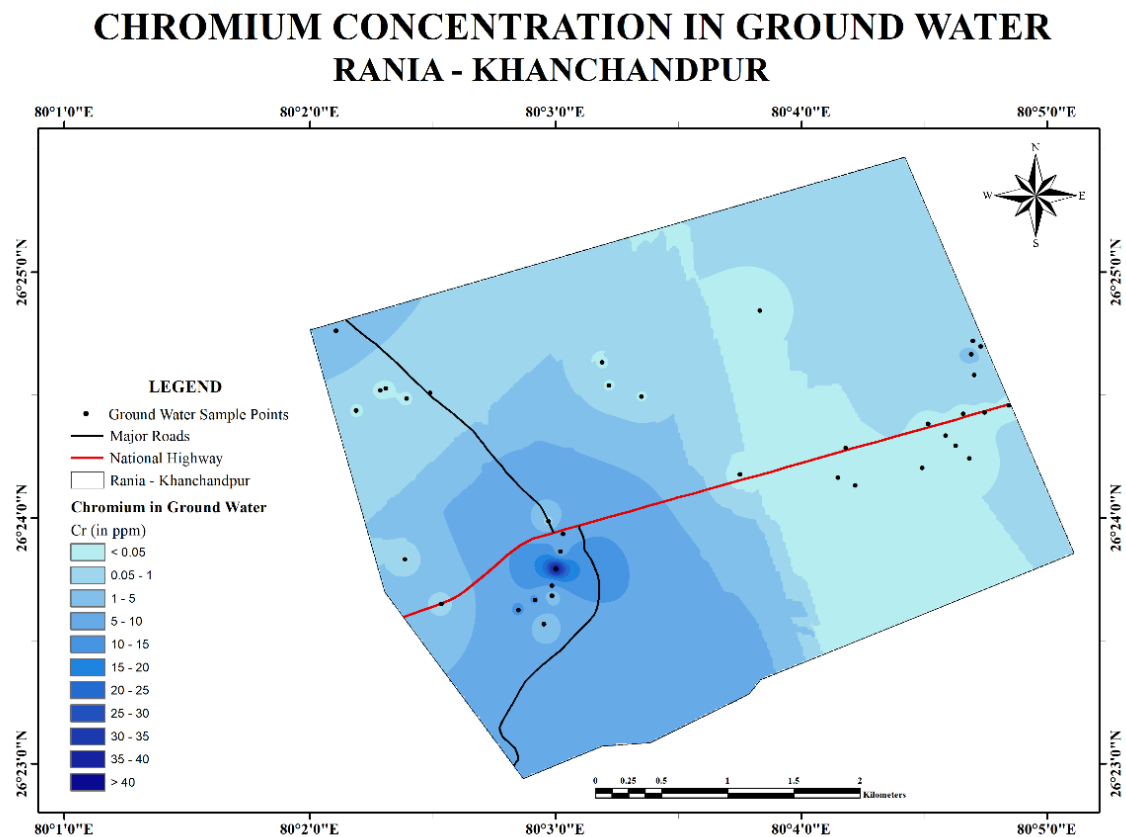


Fig. 7: Spatial Distribution map of chromium contamination in Rania-Khan Chandpur villages in 2022 (Gupta et al., 2024)

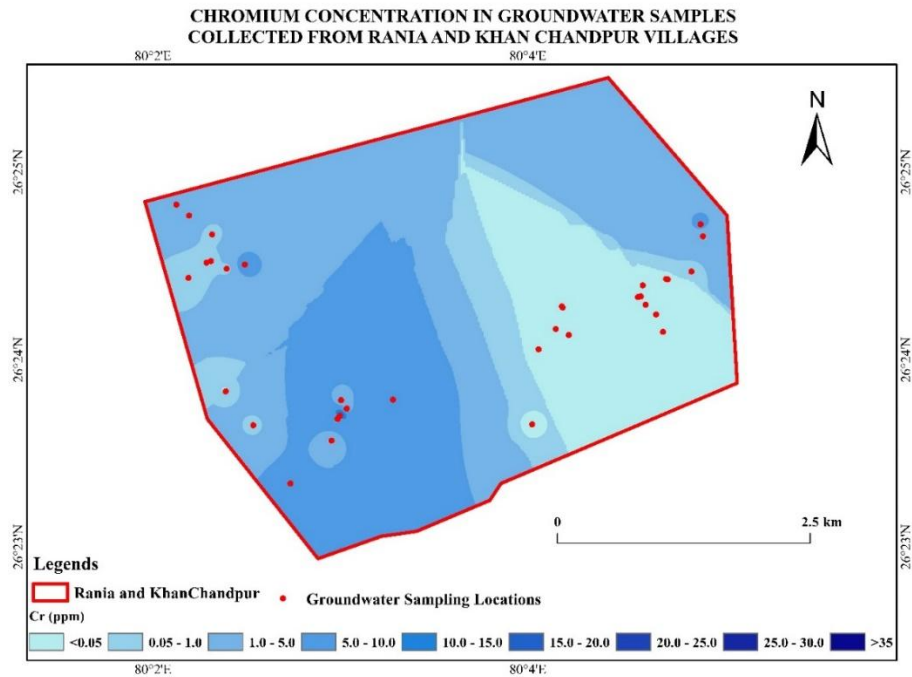


Fig. 8: Spatial Distribution map of chromium contamination in Rania-Khan Chandpur villages in 2023

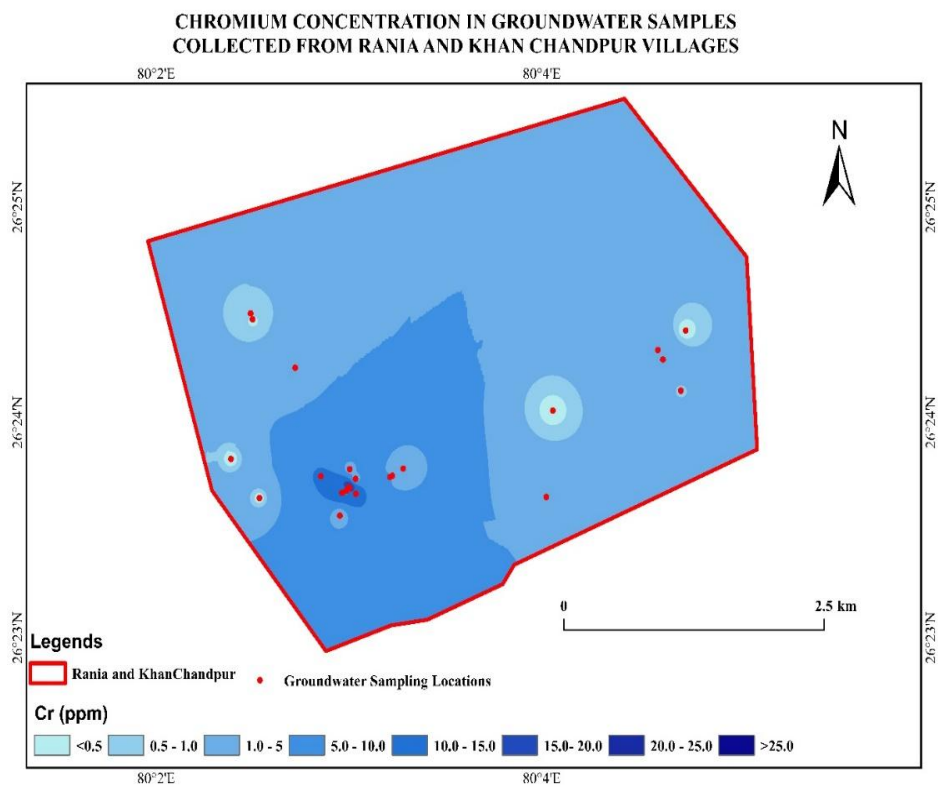


Fig. 9: Spatial Distribution map of chromium contamination in Rania-Khan Chandpur villages in 2024



Fig 10. Prof. Ram Chandra Capacity Building Program in Khan Chandpur Village conducted on 13-15 February 2024 for proper water management in Rania and Khan Chandpur villages.

9.2 Cr contamination in Soil/COPR core at Different Depth

COPR cores were collected from 3 COPR sites (2 from the major COPR site of Rania and 1 from the adjacent COPR site) while 3 soil core samples were collected from in and around the COPR sites in Rania and Khan Chandpur villages. At every place 3 cores have been collected from the surface, 50-55 cm depth, 100-105 cm depth and 150-155 cm depth. Cr concentration in the core increases with increase in depth of the sampling. The maximum Cr (48330 mg Kg^{-1}) was found in the COPR core collected from the main dump site (Fig. 9). In soil samples the maximum Cr reached up to 2918 mg Kg^{-1} .

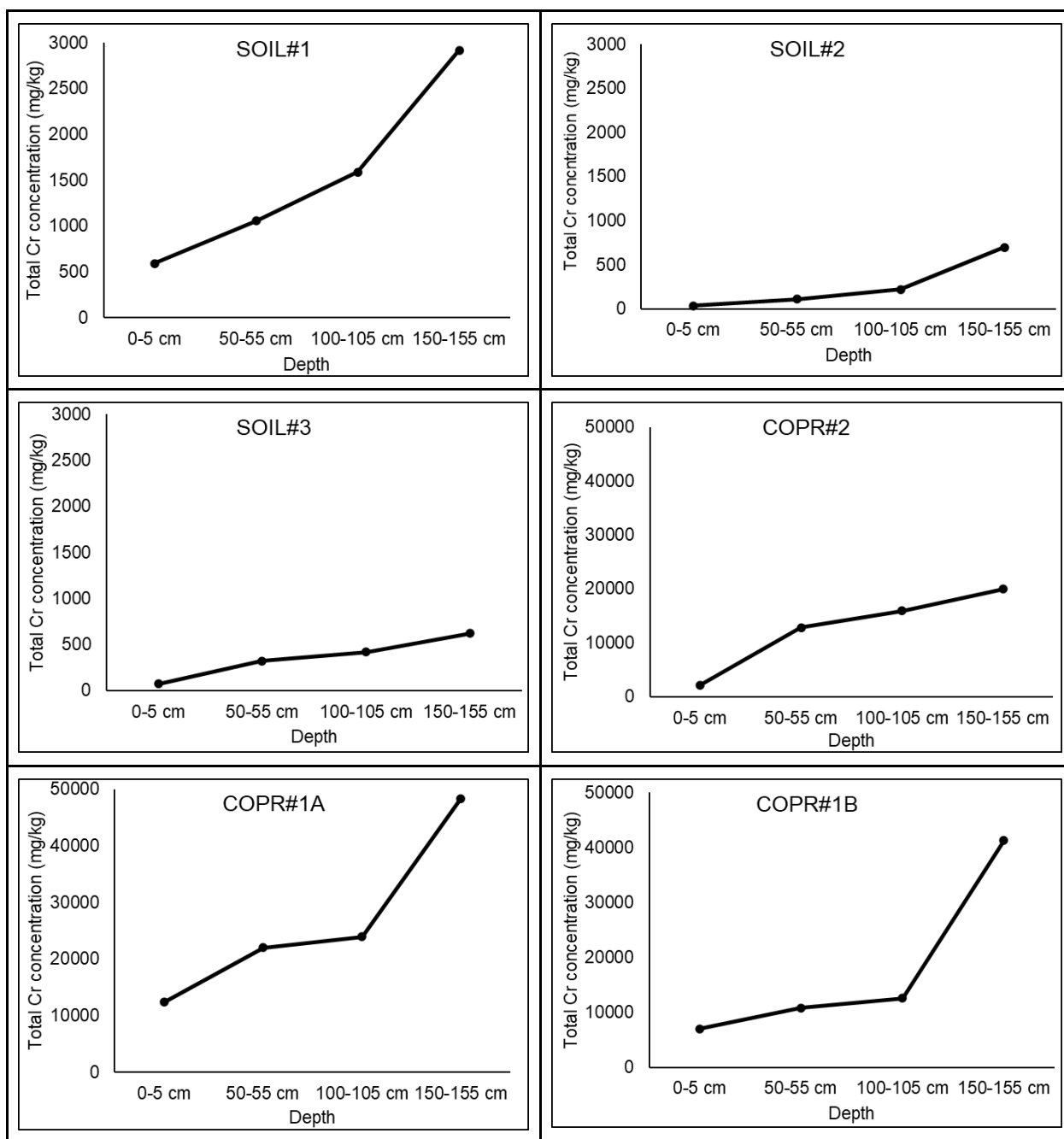


Fig. 9: Cr contamination in Soil and COPR samples collected from Rania and Khan Chandpur villages at different depths

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